

Lesson 22

NATIONAL INCOME DETERMINATION USING DIFFERENTIAL EQUATIONS

TOPIC 089: NATIONAL INCOME DETERMINATION USING DIFFERENTIAL EQUATIONS

If national income model is modelled by following system of equations:

$$Y = C + I$$

$$C = \alpha + \beta Y \quad \alpha > 0, 0 < \beta < 1$$

α is the autonomous consumption and β is the marginal propensity to consume.

$$I = \delta Y + \gamma \left(\frac{dY}{dt} \right) \quad \gamma > 0, \delta < 1$$

Find the time path and the intertemporal equilibrium level of national income.

Solution: Expenditure approach to National income is:

$$Y = \underline{C} + \underline{I}$$

Substituting values of C and I .

$$Y = \alpha + \beta Y + \delta Y + \gamma \left(\frac{dY}{dt} \right)$$

$$Y = \alpha + \beta Y + \delta Y + \gamma \left(\frac{dY}{dt} \right)$$

Rearranging to conform to a first order differential equation:

$$-\gamma \left(\frac{dY}{dt} \right) - \beta Y - \delta Y + Y = \alpha$$

$$-\gamma \left(\frac{dY}{dt} \right) - \beta Y - \delta Y + Y = \alpha$$

$$-\gamma \left(\frac{dY}{dt} \right) + Y - \beta Y - \delta Y = \alpha$$

$$-\gamma \left(\frac{dY}{dt} \right) + (1 - \beta - \delta)Y = \alpha$$

Dividing by $-\gamma$ throughout to normalize the differential equation.

$$\left[-\frac{\gamma}{(-\gamma)} \right] \left(\frac{dY}{dt} \right) \left[+ \right] \frac{(1 - \beta - \delta)}{(-\gamma)} Y = \frac{\alpha}{(-\gamma)}$$

$$\left(\frac{dY}{dt} \right) + \left(\frac{1 - \beta - \delta}{-\gamma} \right) Y = \frac{\alpha}{-\gamma}$$

$$\left(\frac{dY}{dt} \right) + \left(\frac{1 - \beta - \delta}{-\gamma} \right) Y = -\frac{\alpha}{\gamma}$$

Comparing with the standard form of first order differential equation; $\frac{dy}{dt} + ay = b$.

$$\left(\frac{dY}{dt} \right) + \underbrace{\left(\frac{1 - \beta - \delta}{-\gamma} \right)}_a Y = \underbrace{\left(-\frac{\alpha}{\gamma} \right)}_b$$

$$\text{Here } \frac{dy}{dt} = \frac{dY}{dt}, y = Y, a = \left(\frac{1 - \beta - \delta}{-\gamma} \right), b = \left(-\frac{\alpha}{\gamma} \right)$$

Choice of Case of the Differential Equation:

$b = \left(-\frac{\alpha}{\gamma} \right)$; As $\alpha > 0$ and $\gamma > 0$ (both positive constants), their ratio will be non-zero positive constant. However, the value of $b = \left(-\frac{\alpha}{\gamma} \right)$ has a negative sign. The answer will be negative, though still non-zero. i.e. $b \neq 0$ – **Non-Homogenous Case**.

$a = \left(\frac{1-\beta-\delta}{-\gamma}\right)$; Both β and δ are less than 1. The numerator $(1 - \beta - \delta)$ can become 0 only if $\beta + \delta = 1$. So, we assume that the $\beta + \delta \neq 1$ and $(1 - \beta - \delta \neq 0)$, so $\left(\frac{1-\beta-\delta}{-\gamma} \neq 0\right)$ i.e. $a \neq 0$ – **Non-Homogenous Case with $a \neq 0$** .

Therefore, using the definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y}(0) - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a and b , respectively:

$$\underline{Y}(t) = \left\{ \underline{Y}(0) - \frac{\left(\frac{-\alpha}{\gamma}\right)}{\left(\frac{1-\beta-\delta}{-\gamma}\right)} \right\} e^{-\left(\frac{1-\beta-\delta}{-\gamma}\right)t} + \frac{\left(\frac{-\alpha}{\gamma}\right)}{\left(\frac{1-\beta-\delta}{-\gamma}\right)}$$

$$Y(t) = \left\{ Y(0) - \frac{\left(\frac{\alpha}{\gamma}\right)}{\left(\frac{1-\beta-\delta}{-\gamma}\right)} \right\} e^{\left(\frac{1-\beta-\delta}{-\gamma}\right)t} + \frac{\left(\frac{\alpha}{\gamma}\right)}{\left(\frac{1-\beta-\delta}{-\gamma}\right)}$$

$$Y(t) = \left\{ Y(0) - \frac{(\alpha)}{(1-\beta-\delta)} \right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t} + \frac{(\alpha)}{(1-\beta-\delta)}$$

$$Y(t) = \left\{ Y(0) - \left(\frac{\alpha}{1-\beta-\delta}\right) \right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t} + \left(\frac{\alpha}{1-\beta-\delta}\right)$$

Definite Time-path for National Income in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, it should converge towards equilibrium. We know that equilibrium is particular solution (Y_p). Whereas convergence happens when deviation from equilibrium fades to zero. In a differential equation, deviation component is measured by (Y_c). Therefore, for dynamic stability; $Y_c \rightarrow 0$ as $t \rightarrow \infty$.

Using the Definite Time-path for Income level in Dynamic Sense.

$$Y(t) = \underbrace{\left\{ Y(0) - \left(\frac{\alpha}{1-\beta-\delta}\right) \right\}}_{Y_c} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t} + \underbrace{\left(\frac{\alpha}{1-\beta-\delta}\right)}_{Y_p}$$

Need for time path of income $\{Y(t)\}$ to converge to equilibrium income (Y^*) i.e. $Y(t) \rightarrow Y^*$ as $Y_c \rightarrow 0$ when $t \rightarrow \infty$.

$$\begin{aligned} Y_c &\rightarrow 0 \\ \left\{ Y(0) - \left(\frac{\alpha}{1-\beta-\delta}\right) \right\} e^{\left(\frac{1-\beta-\delta}{\gamma}\right)t} &\rightarrow 0 \end{aligned}$$

However, on hair splitting, we don't get the desired result.

$$\left[\underbrace{\left\{ Y(0) - \frac{\left(\frac{\alpha}{1-\beta-\delta}\right)}{\left(\frac{1-\beta-\delta}{-\gamma}\right)} \right\}}_{\substack{Y(0) \geq Y_p \\ \text{Deviation (+ve or -ve)}}} \underbrace{e^{\frac{\left(\frac{1-\beta-\delta}{-\gamma}\right)}{\left(\frac{1-\beta-\delta}{-\gamma}\right)}t}}_{\substack{\text{Exponential} \\ \text{Growth} \\ \text{over time}}} \right] \rightarrow 0$$

Therefore,

$$Y(t) = \left\{ Y(0) - \left(\frac{\alpha}{1 - \beta - \delta} \right) \right\} e^{\left(\frac{1 - \beta - \delta}{\gamma} \right)t} + \left(\frac{\alpha}{1 - \beta - \delta} \right)$$

$$Y(t) = \left\{ Y(0) - \left[\left(\frac{\alpha}{1 - \beta - \delta} \right) \right] \right\} e^{\left(\frac{1 - \beta - \delta}{\gamma} \right)t} + \left[\left(\frac{\alpha}{1 - \beta - \delta} \right) \right]$$

$$Y(t) = \left\{ Y(0) - \left[Y_p \right] \right\} e^{\left(\frac{1 - \beta - \delta}{\gamma} \right)t} + \left[Y_p \right]$$

As particular integral is also the equilibrium value of the variable $[Y_p = Y^*]$.

$$Y(t) = \left\{ Y(0) - [Y^*] \right\} e^{\left(\frac{1 - \beta - \delta}{\gamma} \right)t} + [Y^*]$$

Since addition of (Y^*) in ∞ shall have no effect on infinity;

$$\boxed{Y(t) \rightarrow \infty}$$

$$\boxed{Y(t) \rightarrow Y^*}$$

Interpretation: It can be said that provided $\beta > 0, \delta < 1$ and $\gamma > 0$, regardless of initial national income $Y(0)$ is greater or smaller than equilibrium national income (Y^*), the time-path of national income shall diverge away from equilibrium level of national income as time is allowed to pass indefinitely. Hence the time-path is dynamically unstable.

TOPIC 090: NATIONAL INCOME DETERMINATION: NUMERICAL

Considering a two-sector economy; we have following structural equations:

$$\frac{dY}{dt} = 0.5(C + I - Y)$$

$$C = 0.8Y + 400$$

$$I = 600$$

Find the time-path of national income of the economy; $Y(0) = 7000$.

Solution: Given the equation for rate of national income

$\frac{dY}{dt} = 0.5(C + I - Y)$; we can substitute equations of C and I .

$$\frac{dY}{dt} = 0.5(\underline{C} + \underline{I} - Y)$$

$$\frac{dY}{dt} = 0.5(0.8Y + 400 + 600 - Y)$$

$$\frac{dY}{dt} = 0.5(0.8Y - Y + 400 + 600)$$

$$\frac{dY}{dt} = 0.5(-0.2Y + 1000)$$

$$\frac{dY}{dt} = -0.1Y + 500$$

$$\boxed{\frac{dY}{dt} + 0.1Y = 500}$$

First Order Differential Equation for National Income.

$$\frac{dY}{dt} + \underbrace{0.1Y}_a = \underbrace{500}_b$$

Here $\frac{dy}{dt} = \frac{dY}{dt}$, $y = Y$, $a = 0.1$, $b = 500$, $y(0) = Y(0) = 7000$

Choice of Case of the Differential Equation:

As $b = 500$, so $b \neq 0$: Non-Homogenous Case.

As $a = 0.1$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.

Therefore, using the definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y(0)} - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a , b and $y(0)$ respectively:

$$\underline{Y}(t) = \left(\underline{Y(0)} - \frac{\underline{500}}{\underline{0.1}} \right) e^{-\underline{0.1}t} + \frac{\underline{500}}{\underline{0.1}}$$

$$Y(t) = \left(7000 - \frac{500}{0.1} \right) e^{-0.1t} + \frac{500}{0.1}$$

$$Y(t) = (7000 - 5000)e^{-0.1t} + 5000$$

$$Y(t) = (2000)e^{-0.1t} + 5000$$

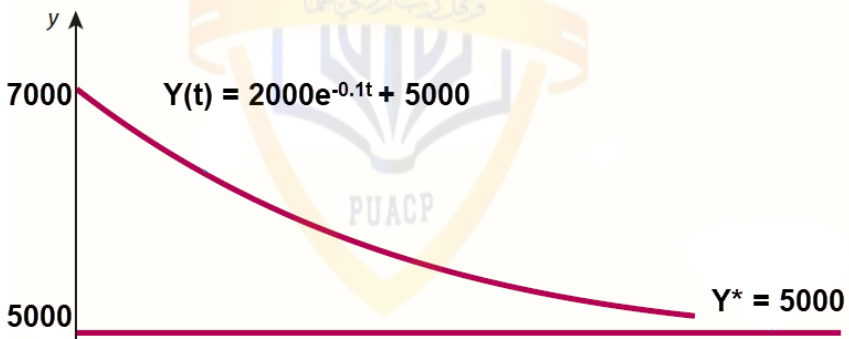
$$\boxed{Y(t) = 2000 e^{-0.1t} + 5000}$$

It is the definite Time-Path of National Income of the economy.

$$Y(t) = \underbrace{2000e^{-0.1t}}_{Y_c} + \underbrace{5000}_{Y_p}$$

Dynamic Stability: The expression; $Y(t) = \boxed{2000e^{-0.1t}} + 5000$ has an exponential decay term in the box. i.e.:

	Complementary Function (Deviation)	Time-Path
t	$Y_c = 2000e^{-0.1t}$	$Y(t) = 2000e^{-0.1t} + 5000$
0	2000	7000
1	1809.675	6809.674836
20	270.6706	5270.670566
40	36.63128	5036.631278
60	4.957504	5004.957504
80	0.670925	5000.670925
100	0.090800	5000.090800
120	0.012288	5000.012288
140	0.001663	5000.001663
160	0.000225	5000.000225
200	0.000004	5000.000004
...	...	...

	Remarks
Positive deviation decays over time	Time-path slides and converges to equilibrium ($Y^* = 5000$)
As $t \rightarrow \infty$, $Y_c \rightarrow 0$	And when $Y_c \rightarrow 0$, $(Y_t) \text{ Converges } (Y^*)$
Graphically	
	
The straight line shows the equilibrium value of national income (Y^*), while curve shows the deviation from equilibrium (Y_c), which decreases over time and leads the time-path to convergence.	

TOPIC 091: INCOME DYNAMICS & INDUCED GOVERNMENT SPENDING USING DIFFERENTIAL EQUATIONS

If national income model is modelled by following system of equations:

$$Y = C + I + G$$

$$C = \alpha + \beta Y \quad \alpha > 0, 0 < \beta$$

α is the autonomous consumption and β is the marginal propensity to consume.

Following is the investment function:

$$I = \delta Y + \gamma \left(\frac{dY}{dt} \right) \quad \gamma > 0, \delta < 1$$

Government Spending is included as the 3rd sector:

$$G = gY \quad g < 1$$

g is the proportion of national income spent as government expenditure.
Find the time path and the intertemporal equilibrium level of national income.

Solution: Expenditure approach to National income is:

$$Y = \underline{C} + \underline{I} + \underline{G}$$

Substituting values of $\underline{C}$, $\underline{I}$ and $\underline{G}$.

$$Y = \underline{\alpha} + \underline{\beta Y} + \underline{\delta Y} + \underline{\gamma \left(\frac{dY}{dt} \right)} + \underline{gY}$$

$$Y = \alpha + \beta Y + \delta Y + \gamma \left(\frac{dY}{dt} \right) + gY$$

Rearranging to conform to a first order differential equation:

$$-\gamma \left(\frac{dY}{dt} \right) + Y - \beta Y - \delta Y - gY = \alpha$$

$$-\gamma \left(\frac{dY}{dt} \right) + (1 - \beta - \delta - g)Y = \alpha$$

Dividing by $-\gamma$ throughout to normalize the differential equation.

$$\left[-\frac{\gamma}{(-\gamma)} \right] \left(\frac{dY}{dt} \right) \left[+ \right] \frac{(1 - \beta - \delta - g)}{(-\gamma)} Y = \frac{\alpha}{(-\gamma)}$$

$$\left(\frac{dY}{dt} \right) + \left(\frac{1 - \beta - \delta - g}{-\gamma} \right) Y = \frac{\alpha}{-\gamma}$$

$$\left(\frac{dY}{dt} \right) + \left(\frac{1 - \beta - \delta - g}{-\gamma} \right) Y = -\frac{\alpha}{\gamma}$$

Comparing with the standard form of first order differential equation; $\frac{dy}{dt} + ay = b$.

$$\left(\frac{dY}{dt} \right) + \underbrace{\left(\frac{1 - \beta - \delta - g}{-\gamma} \right)}_a Y = \underbrace{\left(-\frac{\alpha}{\gamma} \right)}_b$$

$$\text{Here } \frac{dy}{dt} = \frac{dY}{dt}, y = Y, a = \left(\frac{1 - \beta - \delta - g}{-\gamma} \right), b = \left(-\frac{\alpha}{\gamma} \right)$$

Choice of Case of the Differential Equation:

$b = \left(-\frac{\alpha}{\gamma} \right)$; As $\alpha > 0$ and $\gamma > 0$ (both positive constants), their ratio will be non-zero positive constant. However, the value of $b = \left(-\frac{\alpha}{\gamma} \right)$ has a negative sign. The answer will be negative, though still non-zero. i.e. $b \neq 0$ – **Non-Homogenous Case**.

$a = \left(\frac{1 - \beta - \delta - g}{-\gamma} \right)$; the numerator $(1 - \beta - \delta - g)$ can be non-zero if $(\beta + \delta + g) \neq 1$. Also γ is positive value. Therefore, value of a will be some positive/negative (non-zero) value. i.e. $a \neq 0$ – **Non-Homogenous Case with $a \neq 0$** .

Therefore, using the definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y(0)} - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a and b , respectively:

$$\underline{Y}(t) = \left\{ \underline{Y(0)} - \frac{\left[\left(-\frac{\alpha}{\gamma} \right) \right]}{\left(\frac{1 - \beta - \delta - g}{-\gamma} \right)} \right\} e^{-\left[\left(\frac{1 - \beta - \delta - g}{-\gamma} \right) \right] t} + \frac{\left[\left(-\frac{\alpha}{\gamma} \right) \right]}{\left(\frac{1 - \beta - \delta - g}{-\gamma} \right)}$$

$$Y(t) = \left\{ Y(0) - \frac{\left(\frac{\alpha}{1-\beta-\delta-g} \right)}{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)} \right\} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} + \frac{\left(\frac{\alpha}{1-\beta-\delta-g} \right)}{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)}$$

$$Y(t) = \left\{ Y(0) - \frac{(\alpha)}{(1-\beta-\delta-g)} \right\} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} + \frac{(\alpha)}{(1-\beta-\delta-g)}$$

$$Y(t) = \left\{ Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g} \right) \right\} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} + \left(\frac{\alpha}{1-\beta-\delta-g} \right)$$

Define Time-path for National Income in Dynamic Sense. And g has positive effect on the national income, via its negative sign in denominator.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, it should converge towards equilibrium. We know that equilibrium is particular solution (Y_p). Whereas convergence happens when deviation from equilibrium fades to zero. In a differential equation, deviation component is measured by (Y_c). Therefore, for dynamic stability; $Y_c \rightarrow 0$ as $t \rightarrow \infty$.

Using the Define Time-path for Income level in Dynamic Sense.

$$Y(t) = \underbrace{\left\{ Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g} \right) \right\}}_{Y_c} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} + \underbrace{\left(\frac{\alpha}{1-\beta-\delta-g} \right)}_{Y_p}$$

Need for time path of income $\{Y(t)\}$ to converge to equilibrium income (Y^*) i.e. $Y(t) \rightarrow Y^*$ as $Y_c \rightarrow 0$ when $t \rightarrow \infty$.

$$Y_c \rightarrow 0$$

$$\left\{ Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g} \right) \right\} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} \rightarrow 0$$

However, on hair splitting, we don't get the desired result.

$$\left[\underbrace{\left\{ Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g} \right) \right\}}_{\substack{Y(0) \geq Y_p \\ \text{Deviation (+ve or -ve)}}} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} \right] \rightarrow 0$$

Exponential Growth over time

Therefore,

$$Y(t) = \left\{ Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g} \right) \right\} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} + \left(\frac{\alpha}{1-\beta-\delta-g} \right)$$

$$Y(t) = \left\{ Y(0) - \left(\frac{\alpha}{1-\beta-\delta-g} \right) \right\} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} + \left(\frac{\alpha}{1-\beta-\delta-g} \right)$$

$$Y(t) = \left\{ Y(0) - \left[Y_p \right] \right\} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} + \left[Y_p \right]$$

As particular integral is also the equilibrium value of the variable $[Y_p = Y^*]$.

$$Y(t) = \left\{ Y(0) - \left[Y^* \right] \right\} e^{\left(\frac{1-\beta-\delta-g}{1-\beta-\delta-g} \right)t} + \left[Y^* \right]$$

Since addition of (Y^*) in ∞ shall have no effect on infinity;

$$Y(t) \rightarrow \infty$$

$$Y(t) \rightarrow Y^*$$

Interpretation: It can be said that provided $\beta > 0, \delta < 1, \gamma > 0$ and $g < 1$, regardless of initial national income $Y(0)$ is greater or smaller than equilibrium national income (Y^*), the time-path of national income shall **diverge away from equilibrium** level of national income as time is allowed to pass indefinitely. Hence the time-path is **dynamically unstable**.

TOPIC 092: DYNAMICS OF INVESTMENT USING DIFFERENTIAL EQUATIONS

A principal of \$60 is invested. The value, $I(t)$, of the investment, t days later, satisfies the differential equation $\frac{dI}{dt} = 0.002I + 5$. Find the value of the investment if the initial level of investment $y(0)$ is 2000.

Solution: Given the equation for rate of investment $\frac{dI}{dt} = 0.002I + 5$; we can rearrange to compare with the standard form of differential equation.

$$\frac{dI}{dt} = 0.002I + 5$$

$$\frac{dI}{dt} - 0.002I = 5$$

$$\frac{dI}{dt} + \underbrace{(0.002)}_a I = \underbrace{5}_b$$

Here $\frac{dy}{dt} = \frac{dI}{dt}$, $y = I$, $a = 0.002$, $b = 5$, $y(0) = I(0) = 2000$.

Choice of Case of the Differential Equation:

As $b = 5$, so $b \neq 0$: Non-Homogenous Case.

As $a = 0.002$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y(0)} - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a , b and $y(0)$ respectively:

$$\underline{I}(t) = \left(\underline{I(0)} - \frac{\underline{5}}{\underline{0.002}} \right) e^{-\underline{0.002}t} + \frac{\underline{5}}{\underline{0.002}}$$

$$I(t) = \left(2000 - \frac{5}{0.002} \right) e^{-0.002t} + \frac{5}{0.002}$$

$$I(t) = (2000 - 2500)e^{-0.002t} + 2500$$

$$\boxed{I(t) = -500e^{-0.002t} + 2500}$$

It is the definite Time-Path of Investment.

$$I(t) = \underbrace{-500e^{-0.002t}}_{I_c} + \underbrace{2500}_{I_p}$$

Dynamic Stability: The expression; $I(t) = \boxed{-500e^{-0.002t}} + 2500$ has an exponential decay term in the box. i.e.:

	Complementary Function (Deviation)	Time-Path
t	$I_c = -500e^{-0.002t}$	$I(t) = -500e^{-0.002t} + 2500$
0	-500.0000	2000.0000
1	-499.0009	2000.9990
2	-498.0040	2001.9960
3	-497.0089	2002.9910
4	-496.0160	2003.9840
5	-495.0249	2004.9751
6	-494.0359	2005.9641
7	-493.0488	2006.9512
8	-492.0637	2007.9363
9	-491.0805	2008.9194
10	-490.0993	2009.9006
...	...	...

	Remarks
Negative deviation decays over time	Time-path rises and converges to equilibrium ($I^* = 2500$)
As $t \rightarrow \infty$, $I_c \rightarrow 0$	And when $I_c \rightarrow 0$, $(I_t) \rightarrow \overrightarrow{\text{Converges}}(I^*)$
	Graphically
	The straight line shows the equilibrium value of consumption (I^*), while curve shows the deviation from equilibrium (I_c), and this deviation decreases over time and leads the time-path to convergence.

TOPIC 093: DYNAMICS OF INTEREST RATE IN CAPITAL MARKET USING DIFFERENTIAL EQUATIONS

The demand and supply for borrowed capital in the capital market are given by the investment and savings functions, respectively:

$$I = \alpha - \beta i + \rho \frac{di}{dt} \quad S = -\gamma + \delta i + \sigma \frac{di}{dt} \quad (\alpha, \beta, \gamma, \delta)$$

Determine the intertemporal equilibrium level and time path of interest rate $i(t)$. What values of the parameters ρ and σ ensure dynamic stability?

Solution: Equilibrium in the market requires equality of Investment and Savings: ($S = I$).

$$\alpha - \beta i + \rho \frac{di}{dt} = -\gamma + \delta i + \sigma \frac{di}{dt}$$

Rearranging to compare with the standard form of differential equation.

$$-\beta i - \delta i + \rho \frac{di}{dt} - \sigma \frac{di}{dt} = -\alpha - \gamma$$

$$\rho \frac{di}{dt} - \sigma \frac{di}{dt} - \beta i - \delta i = -\alpha - \gamma$$

$$(\rho - \sigma) \frac{di}{dt} - (\beta + \delta) i = -(\alpha + \gamma)$$

Dividing by $(\rho - \sigma)$ for normalizing the differential equation.

$$\frac{(\rho - \sigma) di}{(\rho - \sigma) dt} - \frac{(\beta + \delta)}{(\rho - \sigma)} i = -\frac{(\alpha + \gamma)}{(\rho - \sigma)}$$

$$\frac{di}{dt} - \frac{(\beta + \delta)}{(\rho - \sigma)} i = -\frac{(\alpha + \gamma)}{(\rho - \sigma)}$$

Comparing with the standard form of the differential equation, $\frac{dy}{dt} + ay = b$.

$$\frac{di}{dt} + \left\{ -\frac{(\beta + \delta)}{(\rho - \sigma)} \right\} i = -\frac{(\alpha + \gamma)}{(\rho - \sigma)}$$

$\frac{dy}{dt} \quad a \quad y \quad b$

Here $\frac{dy}{dt} = \frac{di}{dt}$, $y = i$, $a = \left\{ -\frac{(\beta + \delta)}{(\rho - \sigma)} \right\}$, $b = \left\{ -\frac{(\alpha + \gamma)}{(\rho - \sigma)} \right\}$

Choice of Case of the Differential Equation:

$b = \left\{ -\frac{(\alpha + \gamma)}{(\rho - \sigma)} \right\}$; As $\alpha > 0$ and $\gamma > 0$ (both positive constants), their ratio will be non-zero positive constant. However, the value of $b = \left\{ -\frac{(\alpha + \gamma)}{(\rho - \sigma)} \right\}$ whose denominator should be non-zero to avoid undefined answer (∞). The answer will be negative, though still non-zero. i.e. $b \neq 0$ – **Non-Homogenous Case**.

$a = \left\{ -\frac{(\beta + \delta)}{(\rho - \sigma)} \right\}$; Both β and δ are less than 1. The numerator $(\beta + \delta)$ shall be a non-zero positive constant. However, $a = \left\{ -\frac{(\beta + \delta)}{(\rho - \sigma)} \right\}$ so value of numerator $(\rho - \sigma)$ should be non-zero to avoid undefined answer (∞). So $\left[\left\{ -\frac{(\beta + \delta)}{(\rho - \sigma)} \right\} \neq 0 \right]$ i.e. $a \neq 0$ – **Non-Homogenous Case with $a \neq 0$** .

Therefore, using the definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y}(0) - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a and b , respectively:

$$\underline{i}(t) = \left\{ \underline{i}(0) - \frac{\left\{ -\frac{(\alpha + \gamma)}{(\rho - \sigma)} \right\}}{\left\{ -\frac{(\beta + \delta)}{(\rho - \sigma)} \right\}} \right\} e^{-\left\{ -\frac{(\beta + \delta)}{(\rho - \sigma)} \right\}t} + \frac{\left\{ -\frac{(\alpha + \gamma)}{(\rho - \sigma)} \right\}}{\left\{ -\frac{(\beta + \delta)}{(\rho - \sigma)} \right\}}$$

$$\underline{i}(t) = \left[\underline{i}(0) - \frac{\left[\frac{(\alpha + \gamma)}{(\rho - \sigma)} \right]}{\left[\frac{(\beta + \delta)}{(\rho - \sigma)} \right]} \right] e^{\left[\frac{(\beta + \delta)}{(\rho - \sigma)} \right]t} + \frac{\left[\frac{(\alpha + \gamma)}{(\rho - \sigma)} \right]}{\left[\frac{(\beta + \delta)}{(\rho - \sigma)} \right]}$$

$$i(t) = \left[i(0) - \frac{\left\{ \frac{(\alpha + \gamma)}{(\rho - \sigma)} \right\}}{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\}} \right] e^{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} t} + \frac{\left\{ \frac{(\alpha + \gamma)}{(\rho - \sigma)} \right\}}{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\}}$$

$$i(t) = \left\{ i(0) - \frac{(\alpha + \gamma)}{(\beta + \delta)} \right\} e^{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} t} + \frac{(\alpha + \gamma)}{(\beta + \delta)}$$

$$i(t) = \left\{ i(0) - \frac{(\alpha + \gamma)}{(\beta + \delta)} \right\} e^{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} t} + \frac{(\alpha + \gamma)}{(\beta + \delta)}$$

Definite Time-path for Interest rate in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, it should converge towards equilibrium. We know that equilibrium is particular solution (i_p). Whereas convergence happens when deviation from equilibrium fades to zero. In a differential equation, deviation component is measured by (i_c). Therefore, for dynamic stability; $i_c \rightarrow 0$ as $t \rightarrow \infty$.

Using the Definite Time-path for interest rate in Dynamic Sense.

$$i(t) = \underbrace{\left\{ i(0) - \frac{(\alpha + \gamma)}{(\beta + \delta)} \right\}}_{i_c} e^{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} t} + \underbrace{\frac{(\alpha + \gamma)}{(\beta + \delta)}}_{i_p}$$

Need for time path of income $\{i(t)\}$ to converge to equilibrium level of interest rate (i^*) i.e. $i(t) \rightarrow i^*$ as $i_c \rightarrow 0$ when $t \rightarrow \infty$.

$$i_c \rightarrow 0$$

$$\left\{ i(0) - \frac{(\alpha + \gamma)}{(\beta + \delta)} \right\} e^{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} t} \rightarrow 0$$

$\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\}$ is the rate of growth/decay of the exponential expression. Therefore, its negativity shall ensure its decay. i.e. $\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} < 0$. It's already known that β and δ are positive and their sum (Numerator; $\beta + \delta$) will also be positive. So, denominator ($\rho - \sigma$) must be negative. i.e. $(\rho - \sigma) < 0$. In comparative terms, $(\rho < \sigma)$. Therefore, the condition for dynamic stability of the time-path of interest rate $i(t)$ is $(\rho < \sigma)$:

$$\left[\underbrace{\left\{ i(0) - \frac{(\alpha + \gamma)}{(\beta + \delta)} \right\}}_{\substack{\text{Deviation (+ve or -ve)}}} e^{\underbrace{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} t}_{\substack{\text{Exponential} \\ \text{Decay} \\ \text{over time}}}} \right] \rightarrow 0$$

$\{\alpha, \beta, \delta, \gamma > 0\}$ $\{\beta, \delta > 0 \& (\rho < \sigma)\}$

Therefore,

$$i(t) = \underbrace{\left\{ i(0) - \frac{(\alpha + \gamma)}{(\beta + \delta)} \right\}}_{\rightarrow 0} e^{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} t} + \frac{(\alpha + \gamma)}{(\beta + \delta)}$$

$$i(t) = \underbrace{\left\{ i(0) - \left[\frac{(\alpha + \gamma)}{(\beta + \delta)} \right] \right\}}_{\rightarrow 0} e^{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} t} + \left[\frac{(\alpha + \gamma)}{(\beta + \delta)} \right]$$

$$i(t) = \underbrace{\left\{ i(0) - [i_p] \right\}}_{\rightarrow 0} e^{\left\{ \frac{(\beta + \delta)}{(\rho - \sigma)} \right\} t} + [i_p]$$

$$i(t) = \underbrace{\{i(0) - i_p\}}_{\rightarrow 0} e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t} + i_p$$

As particular integral is also the equilibrium value of the variable $i_p = i^*$.

$$i(t) = \underbrace{\{i(0) - i^*\}}_{\rightarrow 0} e^{\left(\frac{\beta+\delta}{\rho-\sigma}\right)t} + i^*$$

$$\boxed{i(t) \rightarrow \infty}$$

Interpretation: It can be said that provided $\beta > 0, \delta > 0$ with $\rho < \sigma$, regardless of initial interest rate $i(0)$ is greater or smaller than equilibrium interest rate (i^*), the time-path of interest rate shall **converge towards equilibrium** level of interest rate as time is allowed to pass indefinitely. Hence the time-path is **dynamically stable**.



Lesson 23

VARIOUS ECONOMIC DYNAMICS USING DIFFERENTIAL EQUATIONS

TOPIC 094: DYNAMICS OF VALUE OF OIL WELL USING DIFFERENTIAL EQUATIONS

The value of an oil well, which is explored, initially is $V(0) = \$2700$, and is decaying at the rate $\frac{dV}{dt} + 0.04 V(t) = 28$.

a) What is the general time path that describes the value of this well?

b) What is the future value of the well at time $t = 10$?

Solution: Given the equation for rate of value of oil-well $\frac{dV}{dt} + 0.04 V(t) = 28$; we can rewrite to compare with the standard form of differential equation:

$$\begin{aligned}\frac{dV}{dt} + 0.04V &= 28 \\ \frac{dV}{dt} + \underbrace{(0.04)}_a \underbrace{V}_y &= \underbrace{28}_b\end{aligned}$$

Here $\frac{dy}{dt} = \frac{dV}{dt}$, $y = V$, $a = 0.04$, $b = 28$, $y(0) = V(0) = 2700$.

Choice of Case of the Differential Equation:

As $b = 28$, so $b \neq 0$: Non-Homogenous Case.

As $a = 0.04$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y(0)} - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a , b and $y(0)$ respectively:

$$\underline{V}(t) = \left\{ \underline{V(0)} - \frac{\underline{28}}{\underline{0.04}} \right\} e^{-\underline{0.04}t} + \frac{\underline{28}}{\underline{0.04}}$$

$$V(t) = \left(2700 - \frac{28}{0.04} \right) e^{-0.04t} + \frac{28}{0.04}$$

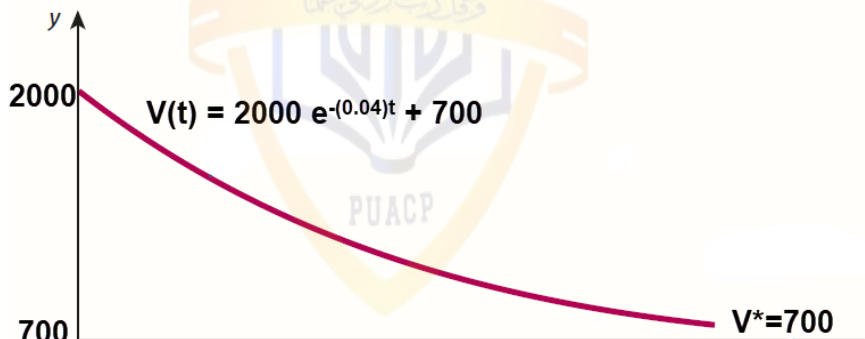
$$V(t) = (2700 - 700)e^{-0.04t} + 700$$

$$\boxed{V(t) = (2000)e^{-0.04t} + 700}$$

It is the definite Time-Path of Value of Oil-Well.

$$V(t) = \underbrace{2000 e^{-0.04t}}_{V_c} + \underbrace{700}_{V_p}$$

Dynamic Stability: The expression; $V(t) = \boxed{2000 e^{-0.04t}} + 700$ has an exponential decay term in the box. i.e.:

	Complementary Function (Deviation)	Time-Path
t	$V_c = 2000 e^{-0.04t}$	$V(t) = 2000 e^{-0.04t} + 700$
0	2000	2700
1	1921.58	2621.58
40	403.793	1103.79
80	81.5244	781.524
120	16.4595	716.459
160	3.32311	703.323
200	0.67093	700.671
240	0.13546	700.135
280	0.02735	700.027
320	0.00552	700.006
340	0.00248	700.002
...	...	...
Remarks		
	Positive deviation decays over time	Time-path slides and converges to equilibrium ($V^* = 700$)
	As $t \rightarrow \infty$, $V_c \rightarrow 0$	And when $V_c \rightarrow 0$, $(V_t) \rightarrow (V^*)$
Graphically		
		
The straight line shows the equilibrium value of Value of Oil-Well (V^*), while curve shows the deviation from equilibrium (V_c), and this deviation decreases over time and leads the time-path to convergence.		

TOPIC 095: DYNAMICS OF VALUE OF GOLD MINE USING DIFFERENTIAL EQUATIONS

The value of a gold mine is initially $G(0) = \$1,400$ and is decaying at the rate $\frac{dG}{dt} + 0.04 G(t) = 16$.

- What is the general time path that describes the value of the gold mine?
- What is the future value of the gold mine at time $t = 10$?

Solution: Given the rate of change of value of Gold mine $\frac{dG}{dt} + 0.04 G(t) = 16$, we can rewrite to compare with the standard form of differential equation:

$$\frac{dG}{dt} + 0.04 G = 16$$

$$\frac{dG}{dt} + \underbrace{(0.04)}_a \underbrace{G}_y = \underbrace{16}_b$$

Here $\frac{dy}{dt} = \frac{dG}{dt}$, $y = G$, $a = 0.04$, $b = 16$, $y(0) = G(0) = 1400$.

Choice of Case of the Differential Equation:

As $b = 16$, so $b \neq 0$: Non-Homogenous Case.

As $a = 0.04$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y(0)} - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a , b and $y(0)$ respectively:

$$\underline{G}(t) = \left\{ \underline{G(0)} - \frac{\underline{16}}{\underline{0.04}} \right\} e^{-\underline{0.04}t} + \frac{\underline{16}}{\underline{0.04}}$$

$$G(t) = \left(1400 - \frac{16}{0.04} \right) e^{-0.04t} + \frac{16}{0.04}$$

$$G(t) = (1400 - 400) e^{-0.04t} + 400$$

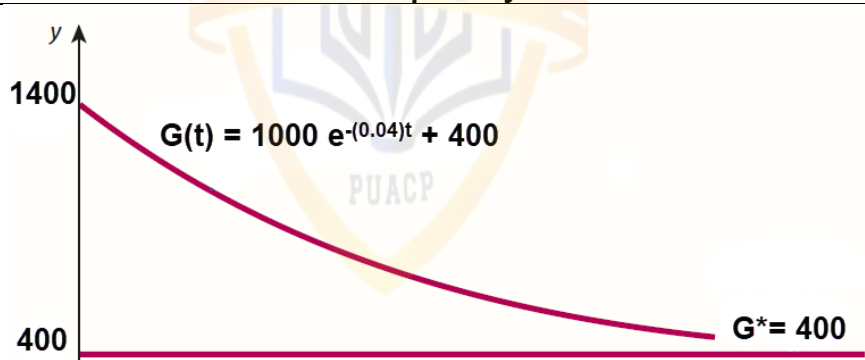
$$\boxed{G(t) = (1000) e^{-0.04t} + 400}$$

It is the definite Time-Path of Value of Gold-Mine.

$$G(t) = \underbrace{1000 e^{-0.04t}}_{G_c} + \underbrace{400}_{G_p}$$

Dynamic Stability: The expression; $G(t) = \boxed{1000 e^{-0.04t}} + 400$ has an exponential decay term in the box. i.e.:

	Complementary Function (Deviation)	Time-Path
t	$G_c = 1000 e^{-0.04t}$	$G(t) = 1000 e^{-0.04t} + 400$
0	1000	1400
1	960.79	1360.789
40	201.90	601.8965
80	40.762	440.7622
120	8.2297	408.2297
160	1.6616	401.6615
200	0.3355	400.3354
240	0.0677	400.0677
280	0.0137	400.0136
320	0.0028	400.0027
360	0.0006	400.0006
...	...	...

	Remarks
Positive deviation decays over time	Time-path slides and converges to equilibrium ($G^* = 400$)
As $t \rightarrow \infty$, $G_c \rightarrow 0$	And when $G_c \rightarrow 0$, $(G_t) \xrightarrow{\text{Converges}} (G^*)$
	Graphically
	
	The straight line shows the equilibrium value of Value of Oil-Well (G^*), while curve shows the deviation from equilibrium (G_c), and this deviation decreases over time and leads the time-path to convergence.

TOPIC 096: HOTELLING'S RULE USING DIFFERENTIAL EQUATIONS

Hotelling's rule is named after statistician and economic theorist Harold Hotelling.

It studies the net revenue of extraction from a nonrenewable natural resource as a function of time $\pi(t)$.

Hotelling found the optimum point to be where net revenue grows exactly at the rate of interest r . Thus, the simple rule is:

$$\frac{\pi'(t)}{\pi(t)} = \underbrace{r}_{\text{rate of interest}}$$

Net revenue

Rewriting the Hotelling's Rule:

$$\frac{\pi'(t)}{\pi(t)} = r$$

$$\pi'(t) = r \times \pi(t)$$

$$\frac{d\{\pi(t)\}}{dt} = r \times \pi(t)$$

In simple terms:

$$\frac{d\pi}{dt} = r\pi$$

Rewriting to compare with standard form:

$$\frac{d\pi}{dt} - r\pi = 0$$

$$\frac{d\pi}{dt} - \underbrace{r}_{a} \underbrace{\pi}_{y} = \underbrace{0}_{b}$$

$$\frac{dy}{dt}$$

$$\text{Here } \frac{dy}{dt} = \frac{d\pi}{dt}, y = \pi, a = r, b = 0$$

Choice of Case of the Differential Equation:

$b = 0$ – Homogenous Case.

$a = r$; interest rate is usually a positive number $r > 0$ i.e. **$a \neq 0$ – Homogenous Case with $a \neq 0$.**

Therefore, using the definite Solution of time-path of **HC**:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \{\underline{y}(0)\}e^{-\underline{a}t}$$

Substituting values of y and a , respectively:

$$\underline{\pi}(t) = \{\underline{\pi}(0)\}e^{-\underline{r}t}$$

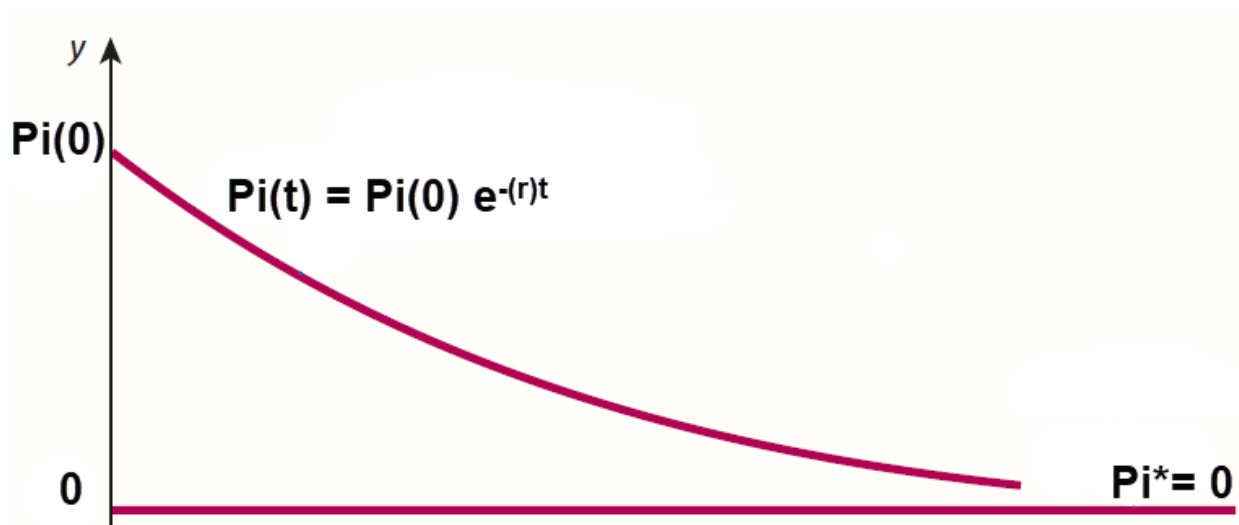
$$\boxed{\pi(t) = \{\pi(0)\}e^{-rt}}$$

Definite Time-path for net revenue in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case (**HC**), the particular solution is zero; $\pi_p = 0$. Therefore, the convergence should take place towards zero (**0**). In this case, we have a negative exponent $\pi(t) = \{\pi(0)\}e^{-rt}$ which shall approach to zero (**0**), when $t \rightarrow \infty$. In a differential equation, deviation component is measured by (π_c) . Therefore, for dynamic stability; $\pi_c \rightarrow 0$ as $t \rightarrow \infty$.

$$\left[\underbrace{\{\pi(0)\}}_{\text{Deviation}} \underbrace{e^{\overbrace{-(r)}^{(r>0)}t}}_{\substack{\text{Exponential} \\ \text{Decay} \\ \text{over time}}} \right] \rightarrow 0$$

Therefore, the time-path of net revenue shall **converge towards equilibrium** as time is allowed to pass indefinitely. Hence the time-path is **dynamically stable**.



TOPIC 097: FUTURE VALUE OF INVESTMENT USING DIFFERENTIAL EQUATIONS

An initial amount $D(0) = \$1,300$ deposited in a bank is growing according to the equation:

$$\frac{dD}{dt} - 0.05 D(t) = 15$$

- What is the general time path that describes the growth of this account?
- What is the future value of this account at time $t = 6$?

Solution: Given the rate of change of value of deposit $\frac{dD}{dt} - 0.05 D(t) = 15$, we can rewrite to compare with the standard form of differential equation:

$$\frac{dD}{dt} + 0.05 D = 15$$

$$\frac{dD}{dt} + \underbrace{(0.05)}_a \underbrace{D}_y = \underbrace{15}_b$$

Here $\frac{dy}{dt} = \frac{dD}{dt}$, $y = D$, $a = 0.05$, $b = 15$, $y(0) = D(0) = 1300$.

Choice of Case of the Differential Equation:

As $b = 15$, so $b \neq 0$: Non-Homogenous Case.

As $a = 0.05$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y(0)} - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a , b and $y(0)$ respectively:

$$\underline{D}(t) = \left\{ \underline{D(0)} - \frac{\underline{15}}{\underline{0.05}} \right\} e^{-\underline{0.05}t} + \frac{\underline{15}}{\underline{0.05}}$$

$$D(t) = (1300 - 300)e^{-0.05t} + 300$$

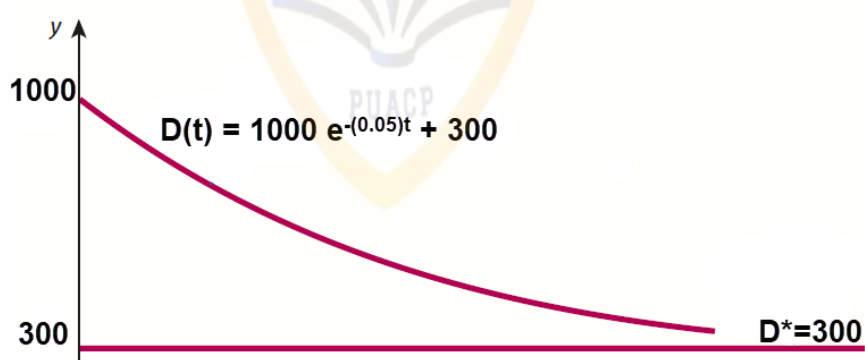
$$D(t) = (1300 - 300)e^{-0.05t} + 300$$

$$\underline{D(t) = (1000)e^{-0.05t} + 300}$$

It is the definite Time-Path of Value of Deposit.

$$D(t) = \underbrace{1000 e^{-0.05t}}_{D_c} + \underbrace{300}_{D_p}$$

Dynamic Stability: The expression; $D(t) = \boxed{1000 e^{-0.05t}} + 300$ has an exponential decay term in the box. i.e.:

	Complementary Function (Deviation)	Time-Path
t	$D_c = \boxed{1000 e^{-0.05t}}$	$D(t) = \boxed{1000 e^{-0.05t}} + 300$
0	1000	1300
1	951.2	1251.22
30	223.1	523.130
60	49.78	349.787
90	11.10	311.108
120	2.478	302.478
150	0.553	300.553
180	0.123	300.123
210	0.027	300.027
240	0.006	300.006
270	0.001	300.001
...	...	...
Remarks		
	Positive deviation decays over time	Time-path slides and converges to equilibrium ($D^* = 400$)
	As $t \rightarrow \infty$, $D_c \rightarrow 0$	And when $D_c \rightarrow 0$, $(D_t) \xrightarrow{\text{Converges}} (D^*)$
Graphically		
		
	The straight line shows the equilibrium value of Value of Deposit (D^*), while curve shows the deviation from equilibrium (D_c), and this deviation decreases over time and leads the time-path to convergence.	

TOPIC 098: DYNAMICS OF MONTHLY SALES USING DIFFERENTIAL EQUATIONS

A model predicts that monthly sales, $S(t)$, of a new product after t months satisfy the differential equation:

$$\frac{dS}{dt} = k(M - S) \text{ with initial condition } S(0) = C.$$

Where k , M and C are positive constants.

Solution: Given the rate of change of monthly sales; $\frac{dS}{dt} = k(M - S)$, we can rewrite it for the sake of comparison with first order differential equation.

$$\begin{aligned}\frac{dS}{dt} &= k(M - S) \\ \frac{dS}{dt} &= kM - kS \\ \frac{dS}{dt} + \underbrace{k}_{a} \underbrace{S}_{y} &= \underbrace{kM}_{b}\end{aligned}$$

Here $\frac{dy}{dt} = \frac{dS}{dt}$, $y = S$, $a = k$, $b = kM$, $y(0) = S(0) = C$

Choice of Case of the Differential Equation:

$b = kM$; As $k > 0$ and $M > 0$ (both positive constants), their product ($k \times M$) will be non-zero positive constant. i.e. $b \neq 0$ – **Non-Homogenous Case**.

$a = k$; $k > 0$ (non-zero positive constants). i.e. $a \neq 0$ – **Non-Homogenous Case with $a \neq 0$** .

Therefore, using the definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y(0)} - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a and b , respectively:

$$\begin{aligned}\underline{S}(t) &= \left\{ \underline{S(0)} - \frac{\underline{kM}}{\underline{k}} \right\} e^{-\underline{k}t} + \frac{\underline{kM}}{\underline{k}} \\ S(t) &= \left\{ \underline{S(0)} - \frac{kM}{k} \right\} e^{-kt} + \frac{kM}{k} \\ S(t) &= (\underline{C} - M)e^{-kt} + M \\ \boxed{S(t) = (C - M)e^{-kt} + M}\end{aligned}$$

Definite Time-path for Monthly Sales in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, it should converge towards equilibrium. We know that equilibrium is particular solution (S_p). Whereas convergence happens when deviation from equilibrium fades to zero. In a differential equation, deviation component is measured by (S_c). Therefore, for dynamic stability; $S_c \rightarrow 0$ as $t \rightarrow \infty$.

Using the Definite Time-path for Income level in Dynamic Sense.

$$S(t) = \underbrace{(C - M)e^{-kt}}_{S_c} + \underbrace{M}_{S_p}$$

Need for time path of Monthly Sales $\{S(t)\}$ to converge to equilibrium income (S^*) i.e. $S(t) \rightarrow S^*$ as $S_c \rightarrow 0$ when $t \rightarrow \infty$.

$$\begin{aligned}& S_c \rightarrow 0 \\ & (C - M)e^{-kt} \rightarrow 0 \\ & \left[\underbrace{\left\{ \overset{\{M>0\}}{C - \hat{M}} \right\}}_{\substack{\text{Deviation} \\ (+ve \text{ or } -ve)}} \underbrace{\overset{\{k>0\}}{e^{-\hat{k}t}}}_{\substack{\text{Exponential} \\ \text{Decay} \\ \text{over time}}} \right] \rightarrow 0\end{aligned}$$

Therefore,

$$\begin{aligned}
 S(t) &= (C - M)e^{-kt} + M \\
 S(t) &= \underbrace{(C - \boxed{M})}_{\rightarrow 0} e^{-kt} + \boxed{M} \\
 S(t) &= \underbrace{(C - \boxed{S_p})}_{\rightarrow 0} e^{-kt} + \boxed{S_p} \\
 \text{As particular integral is also the equilibrium value of the variable } \boxed{S_p = S^*}. \\
 S(t) &= \underbrace{(C - \boxed{S^*})}_{\rightarrow 0} e^{-kt} + \boxed{S^*} \\
 \boxed{S(t) \rightarrow S^*}
 \end{aligned}$$

Interpretation: It can be said that provided $C > 0, k > 0$, regardless of initial national income $S(0)$ is greater or smaller than equilibrium monthly sales (S^*), the time-path of monthly sales shall **converge towards equilibrium** level of monthly sales as time is allowed to pass indefinitely. Hence the time-path is **dynamically stable**.

TOPIC 099: DYNAMICS OF POPULATION USING DIFFERENTIAL EQUATIONS

The population, N , of a small country is currently 5 million. It is expected that this population will grow according to the differential equation.

$$\frac{dN}{dt} = 0.015N$$

where t is measured in years.

Solution: Current population of the country is 5 million. It implies; $N(0) = 5,000,000$.

Given the rate of change of population; $\frac{dN}{dt} = 0.015N$, rewrite to compare with standard form:

$$\begin{aligned}
 \frac{dN}{dt} &= 0.015N \\
 \frac{dN}{dt} - 0.015N &= 0 \\
 \frac{dN}{dt} + (-0.015)N &= 0 \\
 \frac{dN}{dt} + \underbrace{(-0.015)}_a \underbrace{N}_y &= \underbrace{0}_b
 \end{aligned}$$

Here $\frac{dy}{dt} = \frac{dN}{dt}, y = N, a = -0.015, b = 0, y(0) = N(0) = 5000000$

Choice of Case of the Differential Equation:

$b = 0$; – Homogenous Case.

$a = -0.015$; i.e. $a \neq 0$ – Homogenous Case with $a \neq 0$.

Therefore, using the definite Solution of time-path of **HC**:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$\underline{y(t)} = \{\underline{y(0)}\} e^{-\underline{a}t}$$

Substituting values of y and a , respectively:

$$\underline{N(t)} = \{\underline{N(0)}\} e^{-\underline{a}t} = \{\underline{N(0)}\} e^{-\underline{(-0.015)}t}$$

$$\boxed{N(t) = (5000000)e^{0.015t}}$$

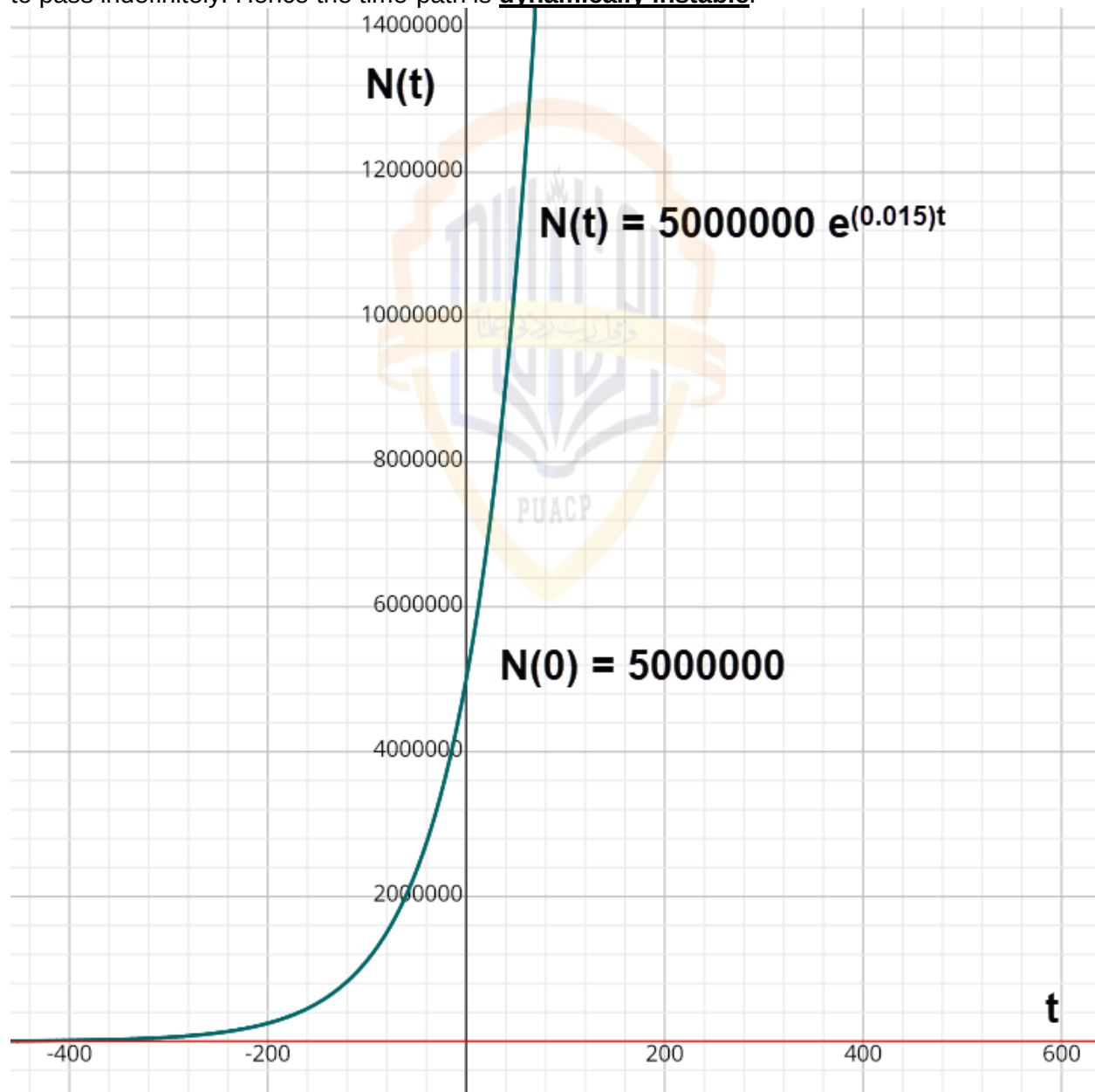
Definite Time-path for net revenue in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case (**HC**), the particular solution is zero; $N_p = 0$. Therefore, the convergence should take place towards zero (**0**). In this case, we have a positive exponent $N(t) = (5000000)e^{0.015t}$ which shall approach to infinity (∞), when $t \rightarrow \infty$. In a differential equation, deviation component is measured by (N_c). Here, there is dynamic instability; $N_c \rightarrow \infty$ as $t \rightarrow \infty$.

$$\left[\underbrace{\{5000000\}}_{\text{Deviation}} \underbrace{e^{(0.015)t}}_{\substack{\text{Exponential} \\ \text{Growth} \\ \text{over time}}} \right] \rightarrow 0$$

($r > 0$)

Therefore, the time-path of population shall diverge away from equilibrium as time is allowed to pass indefinitely. Hence the time-path is dynamically instable.



Policy Problem: From policy perspective, its important to know about the time the population will take to get double. i.e. $\{N(T) = 10,000,000\}_{T=?}$

Using the solution $N(t) = (5000000)e^{0.015t}$ for substitution, we get:

$$\begin{aligned} N(T) &= (5000000)e^{0.015[T]} \\ 10000000 &= (5000000)e^{0.015[T]} \\ \frac{10000000}{5000000} &= e^{0.015[T]} \\ e^{0.015[T]} &= 2 \end{aligned}$$

Raising both sides to exponent:

$$\begin{aligned} \ln(e^{0.015[T]}) &= \ln(2) \\ 0.015[T] &= 0.6931471805599 \\ [T] &= \frac{0.6931471805599}{0.015} \\ T &= 46.2 \end{aligned}$$

Interpretation: It will take **46.2 years** approximately for the population of the country to get double i.e. from 5,000,000 to 10,000,000.



Lesson 24

VARIOUS ECONOMIC DYNAMICS USING DIFFERENTIAL EQUATIONS (CONTINUED)

TOPIC 100: DYNAMICS OF GOVERNMENT SPENDING USING DIFFERENTIAL EQUATIONS

Government expenditure, G , satisfies the differential equation:

$$\frac{dG}{dt} = -0.05G$$

Where, G is measured in billions of dollars and time t is in years. The initial condition is $G(0) = 500$.

(a) Solve this equation to express G in terms of t .

(b) Find the value of G when $t = 6$. Round your answers to three significant figures.

Solution: Given the rate of change of Government Expenditure; $\frac{dG}{dt} = -0.05G$, rewrite to compare with standard form:

$$\begin{aligned}\frac{dG}{dt} &= -0.05G \\ \frac{dG}{dt} + 0.05G &= 0 \\ \frac{dG}{dt} + 0.05G &= 0 \\ \frac{dG}{dt} + \underbrace{(0.05)}_a \underbrace{G}_y &= \underbrace{0}_b\end{aligned}$$

Here $\frac{dy}{dt} = \frac{dG}{dt}$, $y = G$, $a = 0.05$, $b = 0$, $y(0) = G(0) = 500$

Choice of Case of the Differential Equation:

$b = 0$; – Homogenous Case.

$a = 0.05$; i.e. $a \neq 0$ – Homogenous Case with $a \neq 0$.

Therefore, using the definite Solution of time-path of **HC**:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \{\underline{y}(0)\}e^{-\underline{a}t}$$

Substituting values of y and a , respectively:

$$\underline{G}(t) = \{\underline{G}(0)\}e^{-\underline{(0.05)}t}$$

$$G(t) = (500)e^{-0.05t}$$

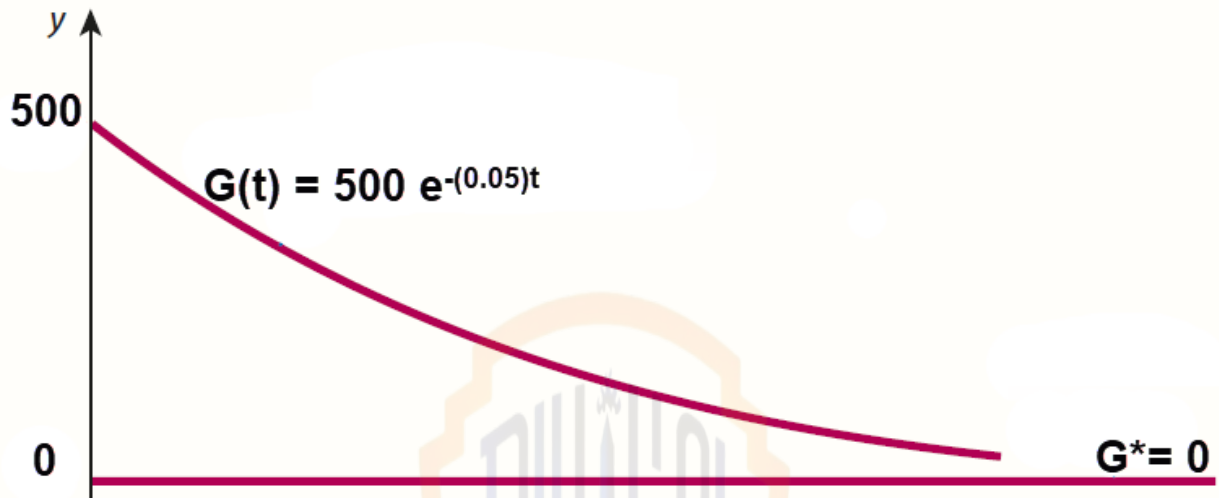
$$\boxed{G(t) = (500)e^{-0.05t}}$$

Definite Time-path for Government Expenditure in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case (**HC**), the particular solution is zero; $G_p = 0$. Therefore, the convergence should take place towards zero (**0**). In this case, we have a negative exponent $G(t) = (500)e^{-0.05t}$ which shall approach to zero (**0**), when $t \rightarrow \infty$. In a differential equation, deviation component is measured by (G_c). Therefore, for dynamic stability; $G_c \rightarrow 0$ as $t \rightarrow \infty$.

$$\left[\underbrace{\{500\}}_{\text{Deviation}} \underbrace{e^{-(0.05)t}}_{\substack{\text{Exponential} \\ \text{Decay} \\ \text{over time}}} \right] \rightarrow 0$$

Therefore, the time-path of Government Expenditure shall converge towards equilibrium as time is allowed to pass indefinitely. Hence the time-path is dynamically stable.



Policy Problem: From policy perspective, it's important to know about the level of government expenditure after some time.

For example; after 6 years: $\{G(6) = ?\}$

Using the solution $G(t) = (500)e^{-0.05t}$ for substitution, we get:

$$G(6) = (500)e^{-0.05[6]}$$

$$G(6) = (500)e^{-0.3}$$

$$G(6) = \$ 370.409 \text{ billion}$$

TOPIC 101: DYNAMICS OF FUEL CONSUMPTION USING DIFFERENTIAL EQUATIONS

Let $F(x)$ denote the number of litre(s) of fuel left in an aircraft's fuel tank if it has flown x km. Suppose that $F(x)$ satisfies the following differential equation: $F'(x) = -aF(x) - b$. Here, the fuel consumption per km is a constant $b > 0$. The term $-aF(x)$, with $a > 0$, is due to the weight of the fuel.

Numerically speaking, the differential equation is: $F'(x) = -0.001F(x) - 8$. The initial level of fuel in the plane is $F(0) = 12000$:

Find the solution of the equation with $F(0) = F_0$.

Solution: Rate of change of remaining fuel; $F'(x) = -0.001F(x) - 8$.

Rewriting equation to compare with the standard form:

$$F'(x) = -0.001F(x) - 8$$

$$F'(x) + 0.001F(x) = -8$$

$$\frac{dF}{dx} + \underbrace{(0.001)}_a \underbrace{F}_y = \underbrace{-8}_b$$

$$\text{Here } \frac{dy}{dt} = \frac{dF}{dx}, y = F, a = 0.001, b = -8, y(0) = F(0) = 12000.$$

Choice of Case of the Differential Equation:As $b = -8$, so $b \neq 0$: Non-Homogenous Case.As $a = 0.001$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.Therefore, using the Definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y(0)} - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a , b and $y(0)$ respectively:

$$\underline{F}(t) = \left\{ \underline{F(0)} - \frac{\underline{(-8)}}{\underline{0.001}} \right\} e^{-\underline{0.001}t} + \frac{\underline{(-8)}}{\underline{0.001}}$$

$$F(t) = \left\{ \underline{12000} - \frac{\underline{(-8)}}{\underline{0.001}} \right\} e^{-(\underline{0.001})t} + \frac{\underline{(-8)}}{\underline{0.001}}$$

$$F(t) = \left\{ \underline{12000} + \frac{\underline{8}}{\underline{0.001}} \right\} e^{-(\underline{0.001})t} - \frac{\underline{8}}{\underline{0.001}}$$

$$F(t) = \{12000 + 8000\}e^{-(0.001)t} - 8000$$

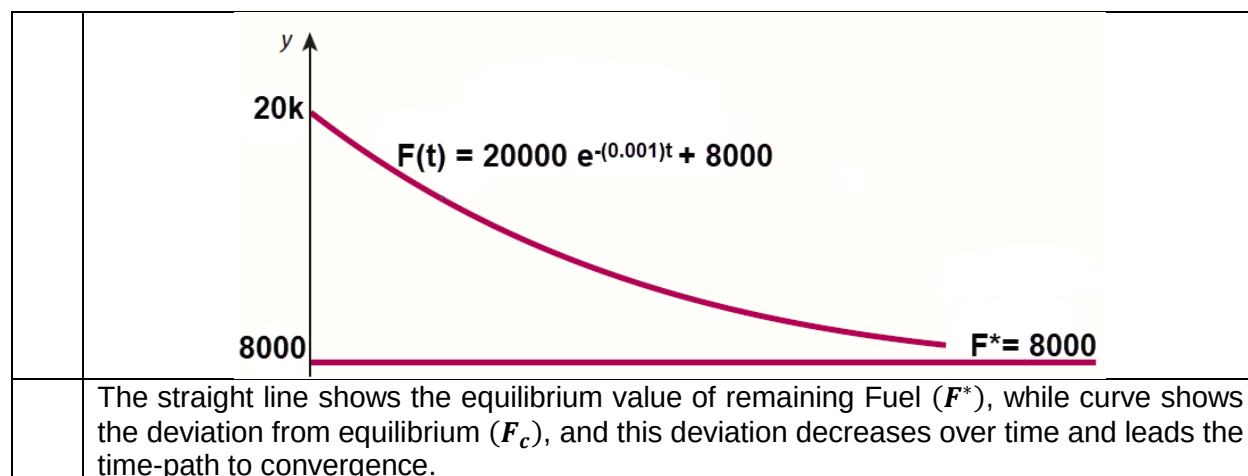
$$F(t) = (20000)e^{-(0.001)t} - 8000$$

It is the definite Time-Path of remaining Fuel.

$$F(t) = \underbrace{20000 e^{-0.001t}}_{F_c} - \underbrace{8000}_{F_p}$$

Dynamic Stability: The expression; $F(t) = \underbrace{20000 e^{-0.001t}}_{F_c} + 8000$ has an exponential decay term in the box. i.e.:

	Complementary Function (Deviation)	Time-Path
t	$F_c = 20000 e^{-0.001t}$	$F(t) = 20000 e^{-0.001t} + 8000$
0	20000	28000
1	19980.0	27980.0
2	19960.0	27960.0
3	19940.0	27940.0
4	19920.1	27920.1
5	19900.2	27900.2
6	19880.3	27880.3
7	19860.4	27860.4
8	19840.6	27840.6
9	19820.8	27820.8
10	19800.9	27800.9
...	...	...
Remarks		
	Positive deviation decays over time	Time-path slides and converges to equilibrium ($F^* = 8000$)
	As $t \rightarrow \infty$, $F_c \rightarrow 0$	And when $F_c \rightarrow 0$, $(F_t) \xrightarrow{\text{Converges}} (F^*)$
Graphically		



TOPIC 102: DYNAMICS OF UNEMPLOYMENT USING DIFFERENTIAL EQUATIONS

If unemployment data is measured in thousands, unemployment in a small economy is measured by following differential equation:

$$\frac{dU}{dt} = -3U + 180$$

Here U is the unemployment level (in thousands) and t is time in years. If unemployment is **150** (in thousands) at the beginning of analysis, find the time-path of unemployment in the economy.

Solution: Given the rate of unemployment in the economy;

$$\frac{dU}{dt} = -3U + 180$$

we can rewrite compare it with the standard form of differential equation.

$$\frac{dU}{dt} + 3U = 180$$

$$\frac{dy}{dt} + \underbrace{3}_{a} \underbrace{U}_{y} = \underbrace{180}_{b}$$

Here $\frac{dy}{dt} = \frac{dU}{dt}$, $y = U$, $a = 3$, $b = 180$, $y(0) = U(0) = 150$.

Choice of Case of the Differential Equation:

As $b = 180$, so $b \neq 0$: Non-Homogenous Case.

As $a = 3$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y(0)} - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a , b and $y(0)$ respectively:

$$\underline{U}(t) = \left\{ \underline{U(0)} - \frac{\underline{180}}{\underline{3}} \right\} e^{-\underline{3}t} + \frac{\underline{180}}{\underline{3}}$$

$$U(t) = \left\{ 150 - \frac{180}{3} \right\} e^{-3t} + \frac{180}{3}$$

$$U(t) = (150 - 60)e^{-3t} + 60$$

$$U(t) = (90)e^{-3t} + 60$$

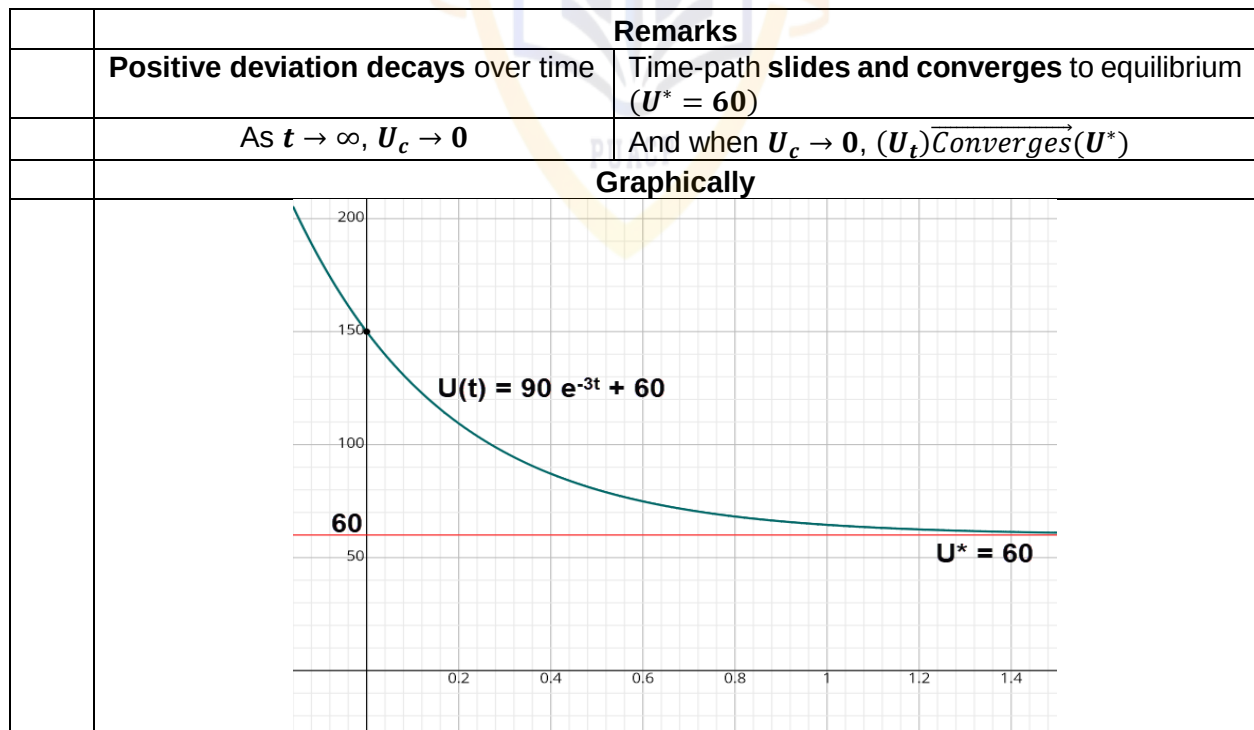
$$\boxed{U(t) = (90)e^{-3t} + 60}$$

It is the definite Time-Path of Unemployment.

$$U(t) = \underbrace{90 e^{-3t}}_{\tilde{U}_c} + \underbrace{60}_{\tilde{U}_p}$$

Dynamic Stability: The expression; $U(t) = \boxed{90 e^{-3t}} + 60$ has an exponential decay term in the box. i.e.:

	Complementary Function (Deviation)	Time-Path
t	$U_c = \boxed{90 e^{-3t}}$	$U(t) = \boxed{90 e^{-3t}} + 60$
0	90	150
1	4.480836153108	64.480836153108
2	0.223087695900	60.223087695900
3	0.011106882368	60.011106882368
4	0.000552979112	60.000552979112
5	0.000027531209	60.000027531209
6	0.000001370698	60.000001370698
7	0.000000068243	60.000000068243
8	0.000000003398	60.000000003398
9	0.000000000169	60.000000000169
10	0.000000000008	60.000000000008
...	...	...

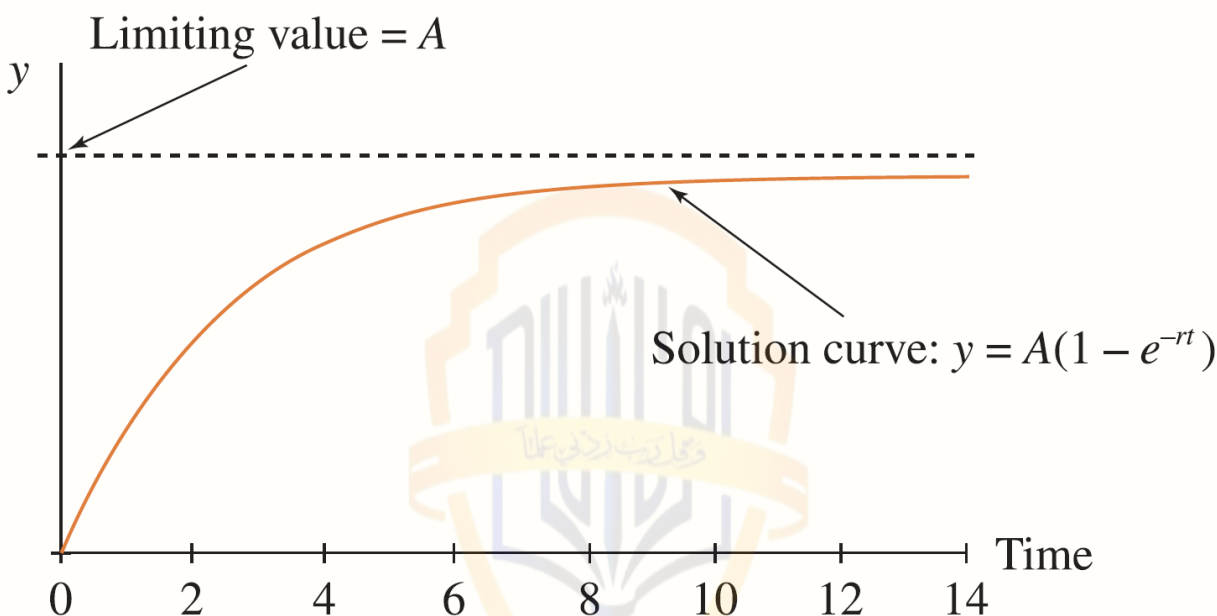


The straight line shows the equilibrium value of Unemployment (U^*), while curve shows the deviation from equilibrium (U_c), and this deviation decreases over time and leads the time-path to convergence.

TOPIC 103: LIMITED GROWTH USING DIFFERENTIAL EQUATIONS

Situations where growth of a variable is fast initially but levels-off (stabilize) just short of a maximum limit. Examples include mobile adoption, wireless internet etc.

Using differential equations of the type $\frac{dy}{dt} = r(A - y)$ represent a similar situation where A is the maximum limit. Graphically it is:



The rate at which the volume of sales (Q) for a new type of printer increases after an advertising campaign is given by the equation $\frac{dQ}{dt} = 0.04(700 - Q)$, given that $Q = 0$ at $t = 0$. Q is the number of printers sold, t is time in years.

(a) Solve the differential equation to obtain an expression for Q in terms of t .

(b) Sketch the solution curve, hence describe verbally how the volume of sales should increase in time.

Solution: Given the rate of change of Volume of Sales; $\frac{dQ}{dt} = 0.04(700 - Q)$, we can rewrite it to compare with standard form of differential equation.

$$\begin{aligned}\frac{dQ}{dt} &= 0.04(700 - Q) \\ \frac{dQ}{dt} &= 28 - 0.04Q \\ \frac{dQ}{dt} + \underbrace{0.04}_{a} \underbrace{Q}_{y} &= \underbrace{28}_{b}\end{aligned}$$

Here $\frac{dy}{dt} = \frac{dQ}{dt}$, $y = Q$, $a = 0.04$, $b = 28$, $y(0) = Q(0) = 0$.

Choice of Case of the Differential Equation:

As $b = 28$, so $b \neq 0$: Non-Homogenous Case.

As $a = 0.04$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.
Therefore, using the Definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y}(0) - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a , b and $y(0)$ respectively:

$$\underline{Q}(t) = \left\{ \underline{Q}(0) - \frac{\underline{28}}{\underline{0.04}} \right\} e^{-\underline{0.04}t} + \frac{\underline{28}}{\underline{0.04}}$$

$$Q(t) = \left\{ 0 - \frac{28}{0.04} \right\} e^{-0.04t} + \frac{28}{0.04}$$

$$Q(t) = (0 - 700)e^{-0.04t} + 700$$

$$Q(t) = (-700)e^{-0.04t} + 700$$

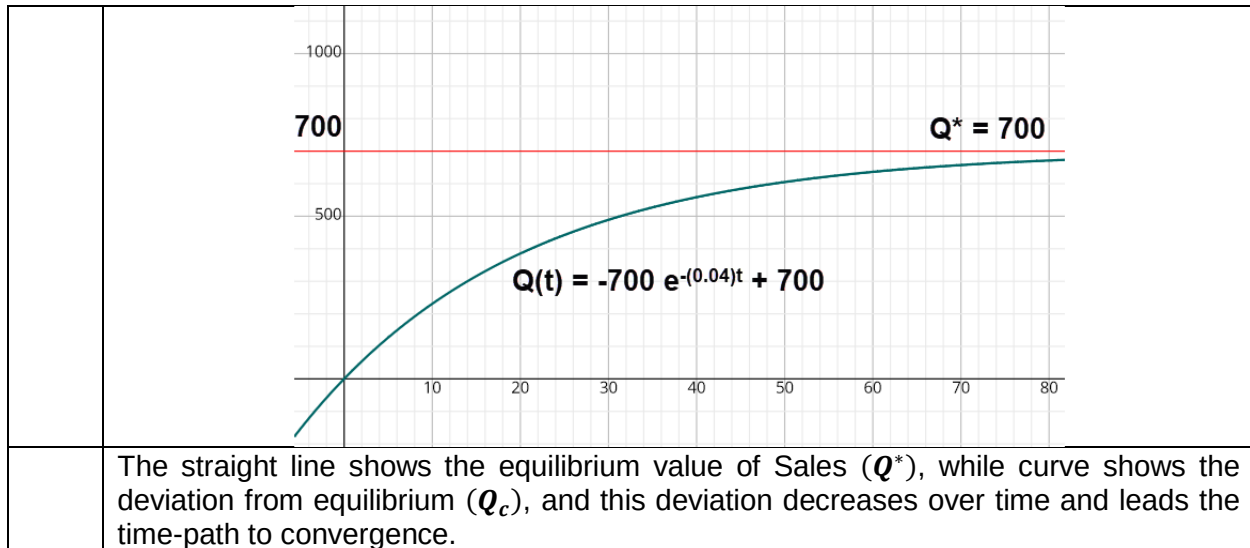
It is the definite Time-Path of Quantity Supplied.

$$Q(t) = \underbrace{-700 e^{-0.04t}}_{Q_c} + \underbrace{700}_{Q_p}$$

Dynamic Stability: The expression; $Q(t) = \underbrace{-700 e^{-0.04t}}_{Q_c} + 700$ has an exponential decay term in the box. i.e.:

	Complementary Function (Deviation)	Time-Path
t	$Q_c = -700 e^{-0.04t}$	$Q(t) = -700 e^{-0.04t} + 700$
0	-700	0
1	-672.552	27.447392
50	-94.7346	605.26530
100	-12.8209	687.17905
150	-1.73512	698.26487
200	-0.23482	699.76517
250	-0.03177	699.96822
300	-0.00430	699.99569
350	-0.00058	699.99941
400	-0.00007	699.99992
450	-0.00001	699.99998
...	...	...

	Remarks	
	Negative deviation decays over time	Time-path rises and converges to equilibrium ($Q^* = 700$)
	As $t \rightarrow \infty$, $Q_c \rightarrow 0$	And when $Q_c \rightarrow 0$, $(Q_t) \xrightarrow{\text{Converges}} (Q^*)$
	Graphically	



TOPIC 104: DYNAMICS OF ORE MINING USING DIFFERENTIAL EQUATIONS

The rate at which an ore is mined is given by the differential equation $\frac{dM}{dt} = 0.02M$. Solve the differential equation, if the total reserves of ore in 1989 were 400 million tons.

Solution: Given the rate of change of Ore Mining ($\frac{dM}{dt} = 0.02M$), we can rearrange it to compare with the standard form of differential equations.

$$\begin{aligned}\frac{dM}{dt} &= 0.02M \\ \frac{dM}{dt} - 0.02M &= 0 \\ \frac{dM}{dt} - \underbrace{0.02}_{a} \underbrace{M}_{y} &= \underbrace{0}_{b}\end{aligned}$$

Here $\frac{dy}{dt} = \frac{dM}{dt}$, $y = M$, $a = -0.02$, $b = 0$, $y(0) = M(0) = 400$

Choice of Case of the Differential Equation:

$b = 0$; – Homogenous Case.

$a = -0.02$; i.e. $a \neq 0$ – Homogenous Case with $a \neq 0$.

Therefore, using the definite Solution of time-path of **HC**:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$y(t) = \{y(0)\}e^{-\underline{a}t}$$

Substituting values of y and a , respectively:

$$\underline{M}(t) = \{M(0)\}e^{\underline{(-0.02)}t}$$

$$\underline{M}(t) = (400)e^{0.02t}$$

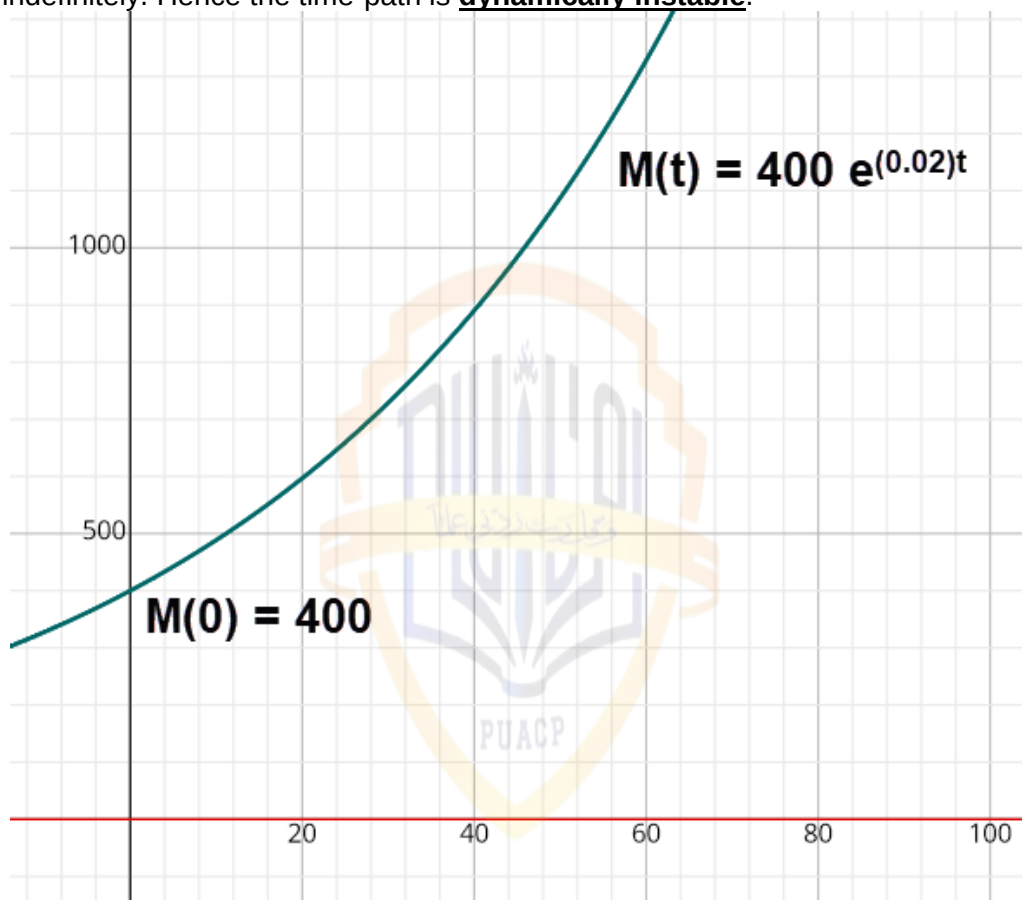
Definite Time-path for Ore Mining in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case (**HC**), the particular solution is zero; $M_p = 0$. Therefore, the convergence should take place towards zero (**0**). In this case, we have a positive exponent $M(t) = (400)e^{0.02t}$ which shall approach to infinity (∞), when $t \rightarrow \infty$. In a

differential equation, deviation component is measured by (M_c). Here, there is dynamic instability; $M_c \rightarrow \infty$ as $t \rightarrow \infty$.

$$\left[\underbrace{\{400\}}_{\text{Deviation}} \underbrace{e^{(0.02)t}}_{\substack{\text{Exponential} \\ \text{Growth} \\ \text{over time}}} \right] \rightarrow 0$$

Therefore, the time-path of Ore Mining shall **diverge away from equilibrium** as time is allowed to pass indefinitely. Hence the time-path is **dynamically instable**.



TOPIC 105: DYNAMICS OF POLLUTANT USING DIFFERENTIAL EQUATIONS

The amount of sulphur dioxide (in tons) present in the atmosphere is given by the equation $\frac{d(P)}{dt} = -0.2P$ (t is in hours). If 50 ton(s) is released into the atmosphere (i.e., $P = 50$ at $t = 0$), derive an expression for the amount of sulphur dioxide present at any time t .

Solution: Given the rate of change of Sulphur Dioxide (SO_2): $\frac{d(P)}{dt} = -0.2P$, we can rewrite it to compare it with the standard form of differential equation.

$$\begin{aligned} \frac{d(P)}{dt} &= -0.2P \\ \frac{dP}{dt} + 0.2P &= 0 \end{aligned}$$

$$\frac{dP}{dt} + \underbrace{0.2}_{\frac{dy}{dt}} \underbrace{P}_{y} = \underbrace{0}_{b}$$

Here $\frac{dy}{dt} = \frac{dP}{dt}$, $y = P$, $a = 0.2$, $b = 0$, $y(0) = P(0) = 50$

Choice of Case of the Differential Equation:

$b = 0$; – Homogenous Case.

$a = 0.2$; i.e. $a \neq 0$ – Homogenous Case with $a \neq 0$.

Therefore, using the definite Solution of time-path of **HC**:

$$y(t) = \{y(0)\}e^{-at}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \{\underline{y}(0)\}e^{-\underline{a}t}$$

Substituting values of y and a , respectively:

$$\underline{P}(t) = \{\underline{P}(0)\}e^{-\underline{(0.2)}t}$$

$$P(t) = (50)e^{-0.2t}$$

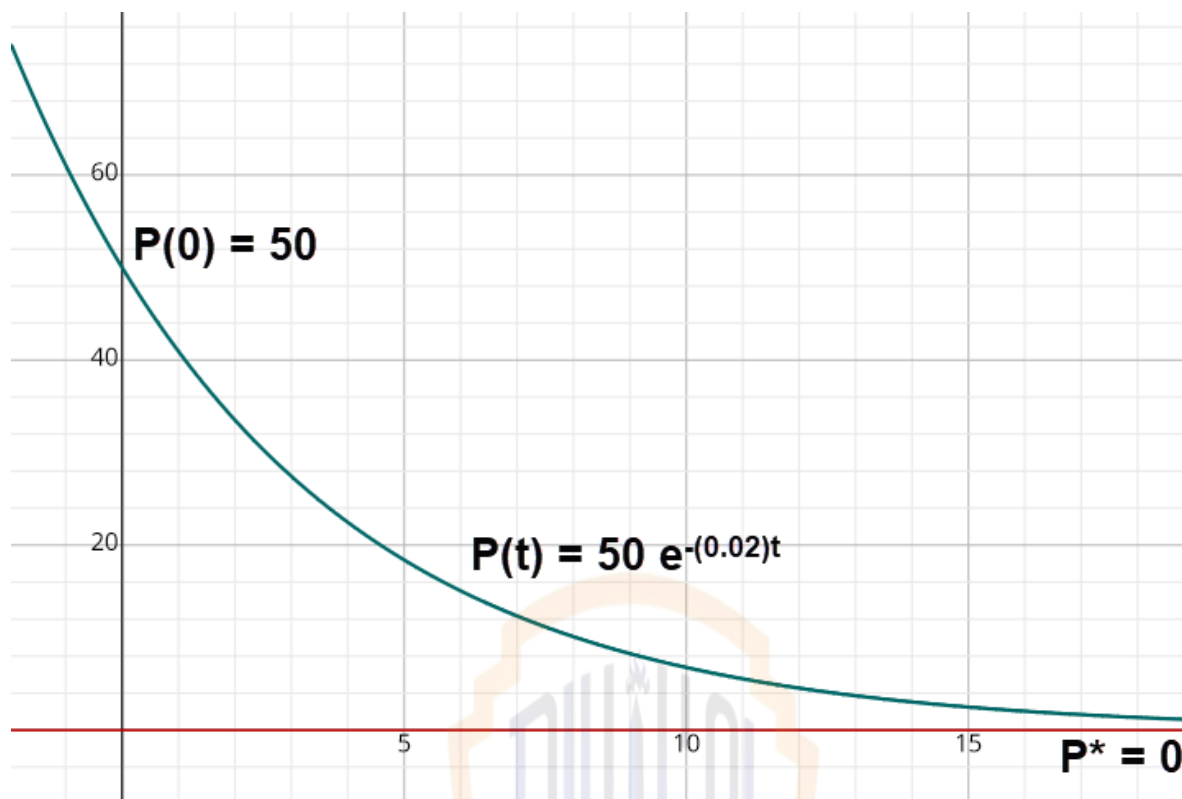
$$P(t) = (50)e^{-0.2t}$$

Definite Time-path for Government Expenditure in Dynamic Sense.

Dynamic Stability: For the dynamic stability of the time-path of First Order Differential Equation, the deviation expression should fade. In homogenous case (**HC**), the particular solution is zero; $P_p = 0$. Therefore, the convergence should take place towards zero (**0**). In this case, we have a negative exponent $P(t) = (50)e^{-0.2t}$ which shall approach to zero (**0**), when $t \rightarrow \infty$. In a differential equation, deviation component is measured by (P_c). Therefore, for dynamic stability; $P_c \rightarrow 0$ as $t \rightarrow \infty$.

$$\left[\underbrace{\{50\}}_{\text{Deviation}} \underbrace{e^{-\underline{(0.2)}t}}_{\substack{\text{Exponential} \\ \text{Decay} \\ \text{over time}}} \right] \rightarrow 0$$

Therefore, the time-path of Pollutant shall converge towards equilibrium as time is allowed to pass indefinitely. Hence the time-path is dynamically stable.



TOPIC 106: DYNAMICS OF FISH STOCK USING DIFFERENTIAL EQUATIONS

The rate at which the fish stocks increase in an inland lake is given by the equation $\frac{dF}{dt} = 0.12(80 - F)$, where F is in thousands, t is in years. (a) Derive an expression for the total number of fish present at any time t given $F = 0$ at $t = 0$.

(b) Graph the total fish stock as a function of time from $t = 0$ to $t = 30$. Does the lake appear to have a maximum carrying capacity?

(c) Calculate the number of years taken for the fish stocks to reach 40000.

Solution: Given the rate of change of fish stock; $\frac{dF}{dt} = 0.12(80 - F)$, we can rearrange it to compare with the standard form of differential equation.

$$\begin{aligned}\frac{dF}{dt} &= 0.12(80 - F) \\ \frac{dF}{dt} &= 9.6 - 0.12F \\ \frac{dF}{dt} + \underbrace{0.12}_{\frac{a}{y}} F &= \underbrace{9.6}_{\frac{b}{a}}\end{aligned}$$

Here $\frac{dy}{dt} = \frac{dF}{dt}$, $y = F$, $a = 0.12$, $b = 9.6$, $y(0) = F(0) = 0$.

Choice of Case of the Differential Equation:

As $b = 9.6$, so $b \neq 0$: Non-Homogenous Case.

As $a = 0.12$, so $a \neq 0$: Non-Homogenous Case with $a \neq 0$.

Therefore, using the Definite Solution of time-path of $NHC_{a \neq 0}$:

$$y(t) = \left\{ y(0) - \frac{b}{a} \right\} e^{-at} + \frac{b}{a}$$

Emphasizing on the values to be substituted:

$$\underline{y}(t) = \left\{ \underline{y}(0) - \frac{\underline{b}}{\underline{a}} \right\} e^{-\underline{a}t} + \frac{\underline{b}}{\underline{a}}$$

Substituting values of y , a , b and $y(0)$ respectively:

$$\underline{F}(t) = \left\{ \underline{F}(0) - \frac{\underline{9.6}}{\underline{0.12}} \right\} e^{-\underline{0.12}t} + \frac{\underline{9.6}}{\underline{0.12}}$$

$$\underline{F}(t) = \left\{ \underline{0} - \frac{\underline{9.6}}{\underline{0.12}} \right\} e^{-\underline{0.12}t} + \frac{\underline{9.6}}{\underline{0.12}}$$

$$\underline{F}(t) = (\underline{0} - \underline{80})e^{-\underline{0.12}t} + \underline{80}$$

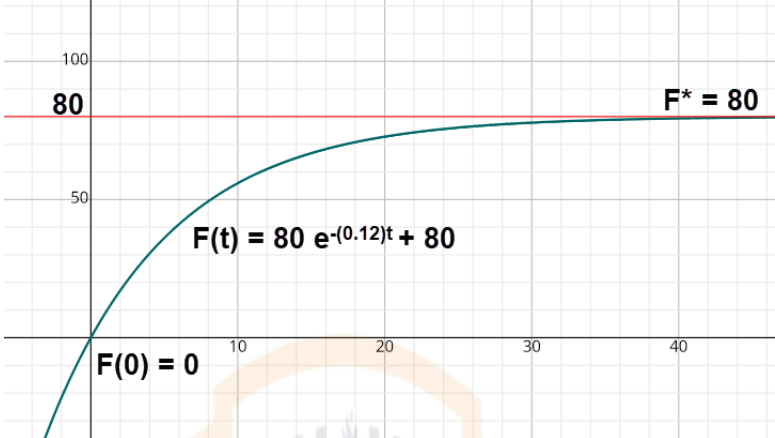
$$\underline{F}(t) = (-\underline{80})e^{-\underline{0.12}t} + \underline{80}$$

It is the definite Time-Path of Fish Stock.

$$\underline{F}(t) = \underbrace{-80 e^{-0.12t}}_{F_c} + \underbrace{80}_{F_p}$$

Dynamic Stability: The expression; $\underline{F}(t) = \underline{-80 e^{-0.12t}} + \underline{80}$ has an exponential decay term in the box. i.e.:

	Complementary Function (Deviation)	Time-Path
t	$F_c = \underline{-80 e^{-0.12t}}$	$\underline{F}(t) = \underline{-80 e^{-0.12t}} + \underline{80}$
0	-80	0
1	-70.95	9.0464
10	-24.10	55.904
20	-7.257	72.743
30	-2.186	77.814
40	-0.658	79.342
50	-0.198	79.802
60	-0.060	79.940
70	-0.018	79.982
80	-0.005	79.995
90	-0.002	79.998
...	...	...

	Remarks	
	Negative deviation decays over time	Time-path rises and converges to equilibrium ($F^* = 80$)
	As $t \rightarrow \infty$, $F_c \rightarrow 0$	And when $F_c \rightarrow 0$, $(F_t) \xrightarrow{\text{Converges}} (F^*)$
	Graphically	
		
	The straight line shows the equilibrium value of Fish Stock (F^*), while curve shows the deviation from equilibrium (F_c), and this deviation decreases over time and leads the time-path to convergence.	